

Predicting Airbnb Revenue

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Figure 1: San Diego Airbnb Listing Image

Abstract

The rise of Airbnb has developed contention for hotels in the short-term rental space. This has created substantial interest in understanding the factors that influence Airbnb listing performance. This study develops a predictive model for estimating Airbnb annual revenue using a large-scale dataset of listings from North America, Europe, Asia-Pacific, and Latin America/Caribbean markets. Data from multiple regional Airbnb datasets were merged and cleaned, with features including property characteristics, host attributes, review metrics, pricing policies, and geographic information. Exploratory data analysis revealed significant variation in revenue across regions, cities, and host types, as well as positive relationships between review quality and listing performance. Multiple machine learning models were evaluated, including linear regression, ridge regression, and random forest regression. The best-performing model was a random forest regressor with 200 trees, achieving a mean absolute error (MAE) of approximately \$10,187 and an R^2 score of 0.560. Results suggest that nonlinear relationships and interactions among listing characteristics play an important role in explaining Airbnb revenue. These findings demonstrate the value of machine learning techniques for revenue prediction and provide insights into factors associated with successful short-term rental listings.

Keywords

Airbnb, Machine Learning, Histogram-based Gradient Boosting Regression, Revenue Prediction

1 Introduction

Since Airbnb has expanded, along with its revenue, using data-driven approaches to find suitable features for revenue is an advantage for property managers. Understanding which factors contribute most strongly to revenue generation is therefore an important problem for both hosts and researchers.

Previous studies have identified a variety of determinants of Airbnb success, including location, property size, review quality, pricing strategies, and host reputation. However, many of these factors interact in complex and potentially nonlinear ways, making traditional statistical approaches insufficient for capturing the full structure of the data. Machine learning methods offer a flexible alternative by identifying patterns across large numbers of features without requiring strong assumptions about functional form.

This study investigates the prediction of Airbnb annual revenue using listing-level data collected from multiple global regions. The objectives are: (1) to explore patterns and relationships among listing characteristics, occupancy, ratings, and revenue, and (2) to evaluate the effectiveness of several machine learning models for revenue prediction. By comparing linear and nonlinear approaches, we aim to determine whether advanced predictive methods provide meaningful improvements in forecasting Airbnb performance.

The repository and demo can be found at <https://github.com/gabetedja/dsc148-project>.

2 Data

2.1 Dataset

The dataset combines Airbnb listings data from four major regions: North America, Europe, Asia-Pacific, and Latin America/Caribbean uploaded by [Jason_AirROI](#) on Kaggle. There are 62 columns and 226,825 rows for this dataset, where each row represents one Airbnb listing. Descriptions for each column can be found at the link.

2.2 Data Wrangling

The datasets were very large and needed to be imported via kaggle-hub, and then a region column was assigned to the corresponding datasets (North America dataset had a North America region column, and so on). Then, the 4 datasets were concatenated together. Additionally, all of the numeric columns were coded as objects, so they had to be made numeric. A function was also created to

change True or False values (case/type sensitive) to 0s and 1s to make them easier to work with. Furthermore, for the bedrooms column, due to a division by zero error, listings with zero bedrooms had the bedroom value replaced with a "1", as 0 can reasonably be interpreted as a studio apartment.

Column	Type	Changes/Creation
photos_count	float64	object to numeric
latitude	float64	object to numeric
longitude	float64	object to numeric
guests	float64	object to numeric
bedrooms	float64	object to numeric
beds	float64	object to numeric
baths	float64	object to numeric
min_nights	float64	object to numeric
cleaning_fee	float64	object to numeric
extra_guest_fee	float64	object to numeric
num_reviews	float64	object to numeric
rating_overall	float64	object to numeric
rating_accuracy	float64	object to numeric
rating_checkin	float64	object to numeric
rating_cleanliness	float64	object to numeric
rating_communication	float64	object to numeric
rating_location	float64	object to numeric
rating_value	float64	object to numeric
amenities_count	int64	extract from amenities
has_pool	int64	extract from amenities
has_hot_tub	int64	extract from amenities
has_parking	int64	extract from amenities
has_wifi	int64	extract from amenities
has_kitchen	int64	extract from amenities
has_washer	int64	extract from amenities
has_dryer	int64	extract from amenities
has_air_conditioning	int64	extract from amenities
has_workspace	int64	extract from amenities
allows_pets	int64	extract from amenities
has_beach_access	int64	extract from amenities
has_gym	int64	extract from amenities
has_elevator	int64	extract from amenities
guests_per_bedroom	float64	guests/bedrooms
beds_per_bedroom	float64	beds/bedrooms
baths_per_guest	float64	baths/guests
log_num_reviews	float64	log(num_reviews)
ratings_x_log_reviews	float64	rating_overall * log_num_reviews
listing_type	str	
room_type	str	
cancellation_policy	str	
country	str	
state	str	
city	str	
Region	str	
professional_management	float64	binary encoding
superhost	float64	binary encoding
ttm_revenue	float64	

Table 1: Columns for Model

Several columns were made to support the model creation, which is also explained in the model section of this paper. Binary boolean columns following the format "has_[amenity]" were used to flag if a listing has certain amenities extracted from the raw amenity

column string. Columns guests_per_bedroom, beds_per_bedroom, and baths_per_guest were made by dividing existing columns. Ultimately, unimportant columns such as the "last 90 days" columns ended up being dropped for the model. The model uses 47 columns as seen in Table 1.

3 Exploratory Data Analysis

The histogram distribution of Airbnb revenues (with extreme outliers removed) is skewed right, with most Airbnbs making under \$18906, which is the edge of the first bin [Figure 2], so we can expect most of our predictions to be on the lower end of the median.

Additionally, Airbnbs typically see the most occupancy during summer, peaking in July, which makes sense with vacation timings [Figure 3].

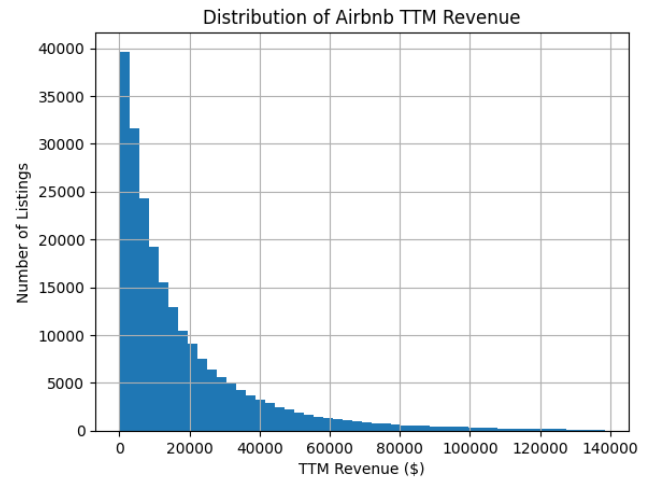


Figure 2: Histogram of Airbnb Listings Revenue

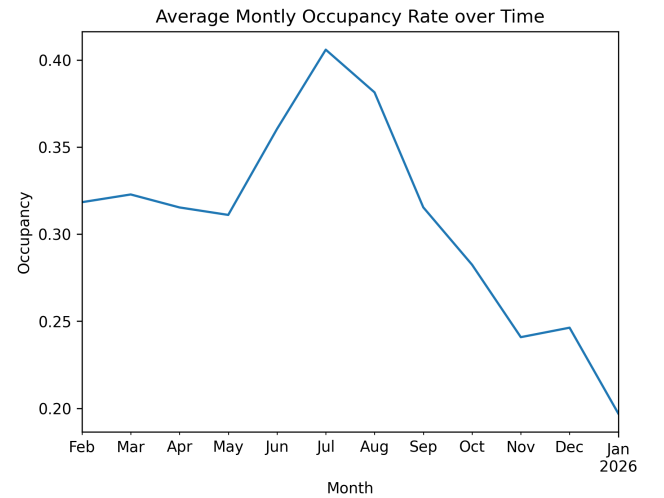


Figure 3: Average Monthly Occupancy Rate Over Months

From the correlation matrix in [Figure 4], we can start to get some insight on which columns are most heavily correlated with revenue, with `ttm_occupancy` having the highest correlation of 0.46. The rest of the columns in the graph have very minimal correlation values with revenue, which could be due to the fact that users can be quite generous with ratings. With the information from [Figure 3], we can likely expect that Airbnbs that are popular for vacation spots have higher occupancy and also higher revenue.

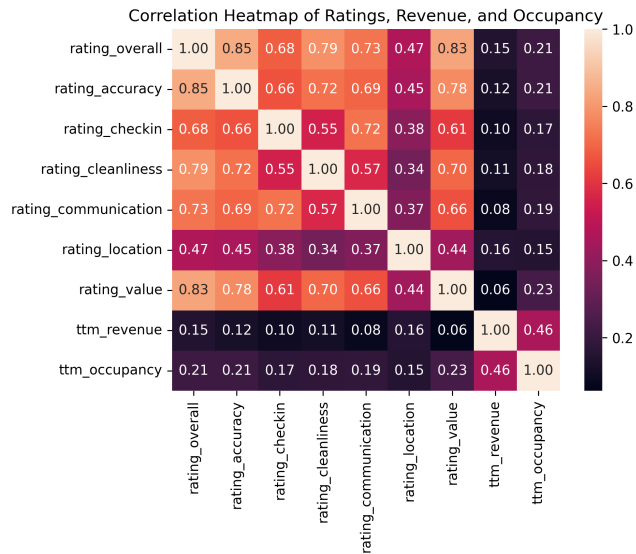


Figure 4: Correlation heatmap of listing ratings, trailing twelve-month revenue, and trailing twelve-month occupancy.

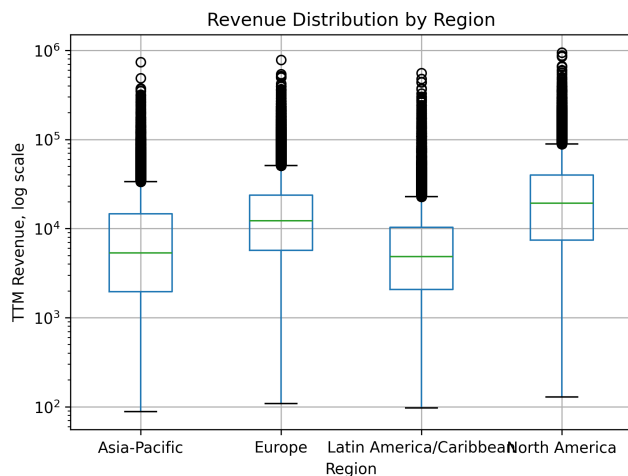


Figure 5: A distribution of Airbnb revenue across North America, Latin America/Caribbean, Asia-Pacific, and Europe

Overall, the revenue quartiles are very similar across all regions, with North America having the highest median revenue on the log

scale [Figure 5]. We can reasonably deduce that occupancy and region (or location) could be strong predictors of revenue, and that `ttm_revenue` can be predicted from different features in the dataset.

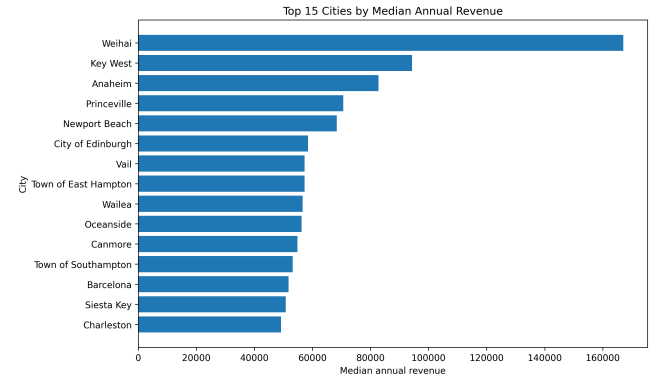


Figure 6: A bar graph showing the cities with the top 15 highest median Airbnb revenues.

The highest revenue cities on average are unsurprisingly popular tourist spots such as Key West, Anaheim, Newport Beach, in which travelers may opt into an Airbnb rather than a hotel. Additionally there's likely less overcompetition in these places compared to say, New York City.

4 Machine Learning Models

4.1 Task Identification

The goal of the model was to forecast the TTM revenue of several Airbnb listings, making it a **regression problem**.

4.2 Evaluation Metric

While RMSE and R^2 were still constantly accounted for within the models, the most important metric for this task is mean absolute error. It answers the most intuitive and useful question: on average, how far off are predictions from the real value? This is very useful for revenue values.

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

RMSE was used to track how our model struggled with outliers, and R^2 was used to see how much variance in revenue is explained by the model.

4.3 Pipeline and Other Constants

To preface, the pipeline used for the model transformers for numeric and categorical features was the same across the iterations. A numeric transformer with median imputation was used. This was meant to fill in missing values like fees and reviews. Median imputation was chosen over mean imputation as mean imputation is more susceptible to outliers than median imputation. In addition, another part of the pipeline is a categorical transformer which is a one hot encoder. The only adjustment made was that for the histogram based gradient regressor, the categorical transformer now had dense output to support the model change. The

models also all used a test train split of 80% training data and 20% test data, with random seed 42.

4.4 Baseline Model

When first working on a model to make predictions on the dataset, the first selected model was a basic linear regression model, as it is simple, fast to execute, and testing for any basic prediction signal at all. While it performed decently, showing there was a signal, it was decidedly too non-linear to perform any better than that. There were obviously non linear reactions making it not optimal for further testing.

4.5 Intermediate Models

Next, a ridge regression was also tried in order not to weigh the smaller sample cities too much, but it did not perform as well as the linear regression model. Ridge regression generally helps with overfitting or multicollinearity, but regularization did not seem to help since the issue was the linear model itself.

Log transforming the model had a lower mean error, but had a significantly worse RMSE and R^2 , so it was quickly disregarded as an option. It reduced the effect of extreme revenue values, helping MAE, but it also had larger errors on expensive properties, explaining why RMSE and R^2 were worse. The relationships were clearly non-linear, and linear regression was mainly just a means to test the model out.

Finally, due to the nature of nonlinear relationships between several variables, the next model tried was a random forest regressor model, with the best results yet. Iterations of this model generally changed the amount of trees used as well as incorporating previously unused features like latitude, longitude, amenities, as well as engineering a new feature, "guests per bedroom". The hyperparameters used in the best model were 500 tree estimators, a maximum depth of 10, a minimum samples leaf of 10, a random seed of 42, and all available CPU cores were used.

Initially, longitude, latitude, and amenities were not included in the models, but it was quickly realized that location and what amenities a given listening has is potentially crucial to revenue.

More features, such as individual amenity flags designating whether a listing has a given amenity, and other ratios like beds per bedroom and baths per guest, as well as log transformed number of reviews, were also added. Typically higher amenity, more spacious listings will generate more revenue than those with low amenities. These additional features yielded marginal improvements.

4.6 Final Model

The final model tried was a HistGradientBoostingRegressor, which had the best mean absolute error, RMSE, and R^2 . It likely performed the best because boosting trains trees sequentially, correcting residuals from the previous trees. Airbnb data is interaction heavy and non-linear, so this made it better suited than the Random Forest regressor, which trains trees independently and thus can fail to correct systematic prediction errors. The hyperparameters in the final model are: max_iter=200, learning_rate=0.1, max_leaf_nodes=63, min_samples_leaf=50, and random_state=42.

4.7 Features Engineered

The improvements made to the model during the iterative process adding amenity count, individual amenity flags (has pool, hot tub, parking, Wi-Fi, air conditioning, kitchen, washer, dryer, workspace, whether or not it allows pets, has gym, has beach access) the amount of guests per bedroom, beds per bedroom, baths per guest, log transformed number of reviews, as well as rating times the log transformed number of reviews to properly weigh sample sizes.

4.8 Feature Importance

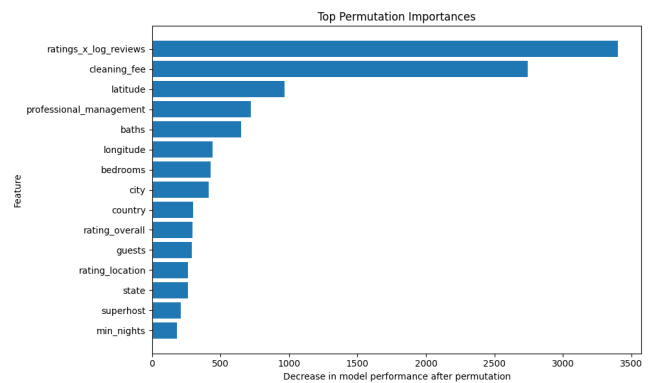


Figure 7: Feature Importance

[Figure 6] ranks features by how much the model's performance worsens if the feature is randomly shuffled. ratings_x_log_reviews has the greatest adverse effect with random shuffling. This makes sense because a listing with a 5.0 rating and only 2 reviews is much less reliable than a 4.8 rating with 300 reviews. cleaning_fee is second highest, likely due to being a stand-in for price tiers or management style (level of luxury). latitude, longitude, city, state, and country are all location features, showing revenue is heavily influenced by geography and market demand. Baths, bedrooms, and guests show that larger listings tend to have a higher revenue potential. Lastly, professional management and superhost show that operation quality of the Airbnb listing matters, with better pricing strategies and better maintenance practice helping guest trust and potentially contribute to greater revenue.

It should be noted that permutation importance does not demonstrate causation, but shows how one variable can forecast another.

5 Literature Review

Many studies have been done on different aspects of Airbnb listings, including occupation, pricing, impact on home values, and more. The studies listed in this review often use different Airbnb datasets, like InsideAirbnb, but we were unable to find published works that used the specific AirROI dataset we use for this model. This is understandable because AirROI is a newer dataset provider, but because all datasets are similarly scraped from Airbnb, we can assume our data to be similar to the ones in this review.

Most studies do not look at specifically predicting revenue, but other aspects that are similarly related. Lektorov, et al. explores

predicting the price of Airbnbs in New York City using a variety of different models including Random Forest, K Neighbors, Support Vector Regression, Decision Trees, Extra Trees and XGB Regressor[1]. They note that decision tree-based models, especially Random Forests, had the lowest error rates and highest determination coefficients. Similarly, Nie’s paper analyzes optimal Airbnb ROI and pricing using XGBoost and specifying what hosts should do to improve ROI[2]. He suggests that location, ratings, and renting out entire rooms increases the ROI for hosts, which offers insight to what increases revenue for certain Airbnbs. Our findings were close to both papers. We found that similar columns were important to revenue generation (4.8 Feature Importance) and that tree-based models performed the best on this task.

The novelty of our work is aimed on predicting revenue instead of a more common task of price, while using a newer dataset from AirROI. We expand on the idea of tree-based models by trying different variations, finalizing with a Histogram-based Gradient Boosting Regressor that explore beyond just Random Forest. With the addition of the ratings_x_log_reviews feature, designed to demonstrate how a high rating is more reliable with a higher amount of reviews, our model is more refined. We also made flags for individual amenities, transforming the vague string into binary values representing the presence of an amenity in a given listing. We additionally made several new capacity ratios that help better represent the spaciousness and/or crowding of a listing. Outside of price, revenue is likely to be of similar or greater interest to Airbnb owners.

6 Results

The results of all our models were as follows.

Model	MAE	RMSE	R ²
Linear Regression	\$11,651.53	\$20,634.59	0.484
Ridge Regression	\$11,723.74	\$20,700.82	0.481
Log-Transformed Ridge Regression	\$9,885.90	\$24,056.75	0.298
Scaled-Down Random Forest	\$10,314.21	\$19,315.55	0.548
Random Forest, 20 Trees	\$10,342.47	\$19,326.24	0.547
Random Forest, 100 Trees	\$10,324.60	\$19,289.30	0.549
Random Forest, 500 Trees	\$10,317.84	\$19,265.52	0.550
Random Forest + Lat/Long/Amenities	\$10,183.96	\$19,060.13	0.560
Random Forest + Guests/Bedroom, 100 Trees	\$10,194.70	\$19,090.92	0.558
Random Forest + Guests/Bedroom, 200 Trees	\$10,186.53	\$19,058.16	0.560
Random Forest + Guests/Bedroom, 500 Trees	\$10,185.03	\$19,061.45	0.560
Random Forest + Ratio Features	\$10,157.69	\$19,077.99	0.559
Random Forest + Amenity Flags	\$10,130.75	\$19,027.14	0.561
Initial HistGradientBoosting	\$9,545.66	\$18,056.97	0.605
Final HistGradientBoosting	\$8,685.41	\$16,630.42	0.665

Table 2: Model Performance Comparison

We can see that the final model, HistGradientBoosting had the best performance metrics.

Demo Page

The demo of this model can be found at this link:

<https://gabetedja.github.io/dsc148-project/>.

Users can fill in inputs for the model, or use the Airbnb webpage scraper which will fill in some of the fields it can find for you. Please note that the site must be hosted for it to work on the web, but can be cloned and hosted locally if needed by changing the API variable. Due to scraping limitations, the scraper functionality does not work when hosted by app services. The result is the estimated TTM Revenue for the user’s inputs.

References

- [1] Alexander Lektorov, Eman Abdelfattah, and Shreehar Joshi. 2023. Airbnb Rental Price Prediction Using Machine Learning Models. In *2023 IEEE 13th Annual Computing and Communication Workshop and Conference (CCWC)*. 0339–0344. doi:10.1109/CCWC57344.2023.10099266
- [2] Jiayang Nie. 2022. Revenue Prediction and Price Optimization - Airbnb. (May 2022). <https://jiayangnie.com/project/airbnb/Airbnb.pdf>